

Responsibility-Oriented Evaluation of Vietnamese Speech Emotion Recognition under Class Imbalance and Silver Labels

Duc Dat Pham^{1,2,4†} , Ngoc Tram Huynh Thi^{3,4†}, Trung Quang Nguyen⁵ ,
Vo Ngoc Minh Anh^{1,4,6}, Le Anh Tien^{7,8}, Tan Duy Le^{*2,4} ,
Kha Tu Huynh^{2,4} , and Nguyen Tan Viet Tuyen⁹

¹ University of Science, Ho Chi Minh City, Vietnam

² High School for the Gifted, Ho Chi Minh City, Vietnam

³ International University, Ho Chi Minh City, Vietnam

⁴ Vietnam National University, Ho Chi Minh City, Vietnam

⁵ Ho Chi Minh City University of Economics and Finance, Ho Chi Minh City, Vietnam

⁶ Ocany Viet Nam Venture CO., LTD, Vietnam

⁷ FPT University, Ho Chi Minh City, Vietnam

⁸ Bot Sales Technology JSC, Vietnam

⁹ School of Electronics and Computer Science, University of Southampton, United Kingdom

[†] These authors contributed equally to this work.

*Corresponding author: ldtan@hcmiu.edu.vn

Abstract. Speech emotion recognition (SER) is increasingly used in intelligent systems, where reliability across emotion categories matters as much as aggregate accuracy. We present a controlled empirical study of Vietnamese SER under class imbalance and silver-label uncertainty using a shared wav2vec2-based backbone. We compare a one-stage flat classifier with a two-stage hierarchical classifier, and examine class-weighted loss, focal loss, minority-aware augmentation, and layer-wise learning-rate decay. Beyond Accuracy and Macro-F1, we evaluate worst-class recall, recall disparity, and stage-wise error decomposition. The best one-stage baseline with standard cross-entropy achieves 75.29% Accuracy and 73.75 Macro-F1, while focal loss with class weights reduces recall disparity from 0.1852 to 0.1327 at the cost of lower Macro-F1 (71.72). In the hierarchical setting, the full fairness-aware recipe improves Stage-2 Macro-F1 from 0.8191 to 0.8439, but degrades end-to-end worst-class recall from 0.6210 to 0.5276 and increases recall disparity from 0.1345 to 0.3834 due to routing errors. These results show that branch-level or fairness-aware gains do not necessarily improve end-to-end reliability, highlighting the value of responsibility-oriented evaluation for Vietnamese SER.

Keywords: Speech Emotion Recognition · Vietnamese SER · Hierarchical Classification · Class Imbalance · Empirical Study

1 Introduction

Speech emotion recognition (SER) is increasingly used in intelligent systems such as mental-health monitoring, affect-aware education, customer-service analytics, and conversational assistants [1]. In these settings, SER outputs may influence downstream recommendations, alerts, and human attention allocation. SER systems should therefore be evaluated not only by predictive performance, but also by how reliably they perform across emotion categories [2].

However, SER is still commonly assessed using aggregate metrics such as accuracy or F1 [3]. This is often insufficient because SER datasets are frequently class-imbalanced and may contain subjective or weakly supervised labels. Under these conditions, strong average performance can hide poor recognition of minority emotions [4]. In addition, standard flat classifiers provide limited visibility into where errors arise, making diagnosis and audit more difficult.

These challenges are especially relevant for Vietnamese SER, which remains less studied than English-centered SER. Existing work has introduced datasets and baseline systems, but has paid less attention to how class imbalance, silver-label uncertainty, and classifier structure jointly affect reliability. In particular, it remains unclear whether fairness-aware training improves minority-class coverage in Vietnamese SER, and whether better branch-level performance in hierarchical models translates into stronger end-to-end behavior.

To address this gap, we present a controlled empirical study of Vietnamese SER under class imbalance and silver-label uncertainty, framing SER as a responsibility oriented decision component whose evaluation should explicitly account for worst-case behavior, class-level disparity, and error attribution across model stages. Rather than proposing a new backbone or loss function, our contribution is a responsibility oriented evaluation framework that jointly analyzes classifier structure, fairness-aware training, and end-to-end reliability under a fixed wav2vec2-based setup. Unlike conventional fairness-aware evaluation, which primarily emphasizes average parity across groups, our framework focuses on end-to-end reliability under structured inference and imperfect supervision. We compare a one-stage flat classifier with a two-stage hierarchical classifier, and examine class-weighted loss, focal loss, minority-aware augmentation, and layer-wise learning-rate decay. Beyond aggregate metrics, we evaluate worst-class recall, recall disparity, and stage-wise error decomposition to provide a more transparent analysis of model behavior.

This paper makes four contributions:

1. **Responsibility-oriented evaluation setting.** We formulate Vietnamese SER under a responsibility-oriented evaluation perspective, where class-level reliability is treated as a primary objective rather than only a secondary diagnostic.
2. **Controlled comparison of classifier structures.** Using the same wav2vec2-based backbone, data split, and training protocol, we compare one-stage and two-stage classifiers to isolate the effect of classifier structure on reliability.

3. **Systematic analysis of fairness-aware training.** We examine class-weighted loss, focal loss, minority-aware augmentation, and layer-wise learning-rate decay, and analyze how they affect both aggregate performance and minority-class coverage.
4. **Practical insight into end-to-end reliability.** We show that improvements in fairness-aware training or branch-level discrimination do not necessarily lead to better end-to-end behavior once routing errors are taken into account.

2 Related Work

Speech emotion recognition (SER) has progressed from handcrafted acoustic pipelines to deep neural architectures that learn representations directly from speech [3]. More recently, pretrained and self-supervised speech encoders, particularly wav2vec 2.0 and related models, have become strong backbones for SER under limited labeled data [5,6]. Despite these advances, most prior work still reports performance mainly through aggregate metrics such as accuracy or F1, offering limited insight into how reliably models behave across emotion categories, especially under imbalance or label uncertainty [3].

These limitations are more pronounced in low-resource and non-English settings. Multilingual SER benchmarks have broadened empirical coverage [7,8], but Vietnamese SER remains relatively underexplored. Existing studies have introduced datasets, baselines, and shared-task results, yet they provide limited analysis of how class imbalance, silver-label uncertainty, and classifier structure interact in practice [9]. In particular, it remains unclear whether techniques that improve branch-level or average performance also improve end-to-end reliability in realistic Vietnamese SER settings.

Class imbalance has long been recognized as a central challenge in classification, motivating approaches such as class weighting, focal loss, and data augmentation [10]. However, these techniques are often evaluated primarily through average performance, rather than through disparity across classes. In this work, we consider fairness at the class level, asking whether emotion recognition quality remains balanced across categories. Different from prior Vietnamese SER studies, we jointly examine imbalance mitigation, hierarchical inference, and silver-label uncertainty under a controlled setup, and evaluate them through a responsibility-oriented perspective centered on worst-class behavior, disparity, and stage-wise error analysis.

3 Methodology

Figure 1 define responsibility-oriented evaluation as a framework that jointly assesses (i) aggregate predictive performance, (ii) worst-case class reliability, and (iii) structural error attribution in multi-stage models. This perspective treats SER systems as decision components whose reliability must be evaluated beyond average-case metrics.

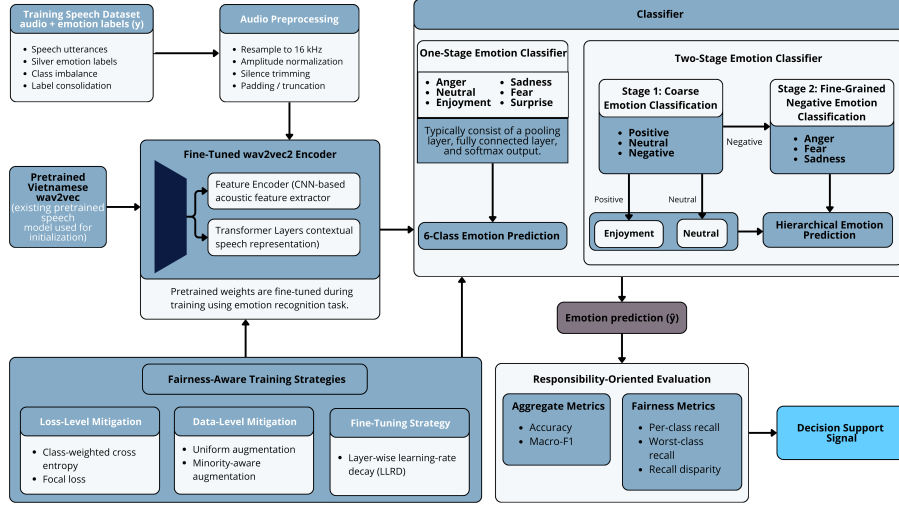


Fig. 1. Overview of the proposed Vietnamese SER pipeline, including wav2vec2-based modeling, reliability-aware training, and responsibility-oriented evaluation.

3.1 Problem Definition and Learning Setting

We study Vietnamese speech emotion recognition (SER) as an utterance-level six-class classification problem under class imbalance and silver-label uncertainty. Let $x \in \mathbb{R}^T$ denote an input waveform, and let the final emotion space be

$$\mathcal{Y} = \{\text{Anger, Enjoyment, Fear, Neutral, Sadness, Surprise}\}, \quad (1)$$

where *Disgust* is merged into *Anger* to reduce sparsity. Since the labels are externally assigned rather than manually verified as gold annotations, we treat the dataset as a silver-label corpus. All systems use the same fixed stratified train/validation/test split. This design isolates the effects of classifier structure and training objectives from variation caused by data partitioning.

3.2 Responsibility-Aware Hierarchical Inference

We use the same pretrained Vietnamese wav2vec 2.0 encoder as the shared acoustic backbone for all systems, so that the comparison focuses on inference structure and optimization rather than representation choice.

One-stage baseline. The one-stage classifier directly predicts a posterior over the six-class label space:

$$p_\psi(y | x), \quad y \in \mathcal{Y}. \quad (2)$$

Two-stage hierarchical classifier. To improve auditability and isolate where failures arise, we decompose SER into coarse emotion routing followed by fine-grained negative-emotion discrimination. Stage 1 predicts

$$q_\theta(z \mid x), \quad z \in \{\text{Positive}, \text{Neutral}, \text{Negative}\}, \quad (3)$$

and Stage 2 predicts

$$r_\phi(c \mid x, \text{Negative}), \quad c \in \mathcal{Y}_-, \quad (4)$$

where $\mathcal{Y}_- = \{\text{Anger}, \text{Fear}, \text{Sadness}, \text{Surprise}\}$.

Instead of relying only on hard routing, we define a structured posterior over the final six-class space:

$$P(y = c \mid x) = \begin{cases} q_\theta(\text{Positive} \mid x), & c = \text{Enjoyment}, \\ q_\theta(\text{Neutral} \mid x), & c = \text{Neutral}, \\ q_\theta(\text{Negative} \mid x) r_\phi(c \mid x, \text{Negative}), & c \in \mathcal{Y}_-. \end{cases} \quad (5)$$

The final prediction is

$$\hat{y} = \arg \max_{c \in \mathcal{Y}} P(y = c \mid x). \quad (6)$$

This soft hierarchical inference makes the model less brittle than hard routing, since uncertainty in Stage 1 is propagated rather than converted immediately into an irreversible decision. It also gives the hierarchical system a transparent probabilistic interpretation: a negative-emotion prediction is reliable only when both coarse routing and within-branch discrimination are reliable.

To stabilize Stage 2, the encoder of the negative branch is initialized from the trained Stage-1 encoder:

$$\phi_{\text{enc}}^{(0)} \leftarrow \theta_{\text{enc}}^{(*)}, \quad (7)$$

where $\theta_{\text{enc}}^{(*)}$ denotes the optimized Stage-1 encoder parameters.

3.3 Reliability-Aware Training Objective

Our training objective is designed to address three practical challenges simultaneously: class imbalance, routing reliability, and silver-label uncertainty.

Class-balanced classification loss. Let n_c be the number of training examples in class c , and let $n_{\max} = \max_c n_c$. We define the clipped inverse-frequency class weight as

$$\alpha_c = \min\left(\frac{n_{\max}}{n_c}, \alpha_{\max}\right). \quad (8)$$

For a classifier head h , with predicted probability $p_y^{(h)}$ for the target class y , we use either class-weighted cross-entropy or class-weighted focal loss:

$$\mathcal{L}_{\text{cls}}^{(h)}(x, y) = \begin{cases} -\alpha_y \log p_y^{(h)}, & \text{weighted cross-entropy,} \\ -\alpha_y (1 - p_y^{(h)})^\gamma \log p_y^{(h)}, & \text{weighted focal loss.} \end{cases} \quad (9)$$

Silver-aware target smoothing. Because the labels are silver rather than fully verified gold annotations, we replace the one-hot target with a softened target distribution:

$$\tilde{\mathbf{y}}_c = (1 - \varepsilon_c)\mathbf{e}_c + \frac{\varepsilon_c}{|\mathcal{Y}|}\mathbf{1}, \quad (10)$$

where $\mathbf{e}_c$ is the one-hot vector for class c . To reflect both label uncertainty and class rarity, the smoothing factor is defined as

$$\varepsilon_c = \text{clip}(\varepsilon_0(1 + \beta \log r_c), \varepsilon_{\min}, \varepsilon_{\max}), \quad r_c = \frac{n_{\max}}{n_c}. \quad (11)$$

This design makes the model less overconfident on sparse classes whose labels are also more likely to be noisy.

Routing-aware regularization. In the hierarchical model, end-to-end failure often originates from incorrect Stage-1 routing. To improve routing reliability for negative emotions, we introduce a margin-based routing loss:

$$\mathcal{L}_{\text{route}}(x, y) = \mathbf{1}[y \in \mathcal{Y}_-] \max\left(0, m - (q_\theta(\text{Negative} \mid x) - \max(q_\theta(\text{Positive} \mid x), q_\theta(\text{Neutral} \mid x)))\right), \quad (12)$$

where $m > 0$ is a margin. This term encourages negative samples to be routed to the negative branch with sufficient confidence, rather than merely winning by a small probability gap.

Unified objective. For the one-stage model, the training loss is

$$\mathcal{L}_{\text{flat}} = \mathcal{L}_{\text{cls}}^{(\text{flat})}(\tilde{\mathbf{y}}). \quad (13)$$

For the hierarchical model, the total loss is

$$\mathcal{L}_{\text{hier}} = \mathcal{L}_{\text{cls}}^{(1)}(\tilde{\mathbf{y}}^{(1)}) + \lambda_{\text{br}}\mathbf{1}[y \in \mathcal{Y}_-]\mathcal{L}_{\text{cls}}^{(2)}(\tilde{\mathbf{y}}^{(2)}) + \lambda_{\text{rt}}\mathcal{L}_{\text{route}}, \quad (14)$$

where $\tilde{\mathbf{y}}^{(1)}$ and $\tilde{\mathbf{y}}^{(2)}$ denote the smoothed targets for Stage 1 and Stage 2, respectively. This objective jointly optimizes coarse routing, fine-grained negative discrimination, and routing reliability.

3.4 Imbalance-Aware Optimization and Augmentation

Minority-aware augmentation. To reduce under-exposure of minority classes, we use class-conditioned waveform augmentation. Given the rarity ratio $r_c = n_{\max}/n_c$, the augmentation probability for class c is

$$\rho_c = \text{clip}\left(p_{\min} + (p_{\max} - p_{\min}) \min\left(1, \frac{1}{2} \log_{10}(r_c) + \frac{1}{2}\right), p_{\min}, p_{\max}\right). \quad (15)$$

This schedule increases perturbation probability as the class becomes rarer, while keeping augmentation bounded for training stability. The perturbation pool includes gain adjustment, additive noise, pitch shifting, time stretching, and time masking.

Layer-wise learning-rate decay. To preserve lower-level acoustic structure while allowing upper layers to adapt more aggressively, we apply layer-wise learning-rate decay (LLRD). Let L be the top encoder layer and let $\ell \in \{0, \dots, L\}$ index a given layer. The learning rate of layer ℓ is

$$\eta_\ell = \eta_0 \lambda^{L-\ell}, \quad 0 < \lambda < 1. \quad (16)$$

Thus, upper layers receive larger updates and lower layers are changed more conservatively.

3.5 Responsibility-Oriented Evaluation

We separate aggregate predictive quality from class-level reliability and hierarchical auditability. For all systems, we report Accuracy, Macro-F1, Macro-Precision, Macro-Recall, per-class recall, and confusion matrices.

Class-level reliability. Let R_c denote the recall of class $c \in \mathcal{Y}$. We quantify worst-case behavior through

$$R_{\text{worst}} = \min_{c \in \mathcal{Y}} R_c, \quad (17)$$

and class disparity through

$$\Delta_{\text{rec}} = \max_{c \in \mathcal{Y}} R_c - \min_{c \in \mathcal{Y}} R_c. \quad (18)$$

A model is preferred not only when its mean performance is high, but also when its weakest class is not systematically neglected.

Stage-wise error attribution. For the hierarchical model, we additionally decompose end-to-end recall for each negative emotion $c \in \mathcal{Y}_-$. Define

$$\begin{aligned} R_{c,\text{route}} &= P(\hat{g}(x) = \text{Negative} \mid y = c), \\ R_{c,\text{branch}} &= P(\hat{y}^{(2)}(x) = c \mid y = c, \hat{g}(x) = \text{Negative}), \end{aligned} \quad (19)$$

where $\hat{g}(x)$ is the Stage-1 routing decision and $\hat{y}^{(2)}(x)$ is the Stage-2 branch prediction. The end-to-end recall of class c factorizes as

$$R_c^{\text{E2E}} = R_{c,\text{route}} \cdot R_{c,\text{branch}}. \quad (20)$$

Equivalently, the miss probability decomposes into

$$1 - R_c^{\text{E2E}} = (1 - R_{c,\text{route}}) + R_{c,\text{route}}(1 - R_{c,\text{branch}}). \quad (21)$$

This decomposition makes it possible to distinguish whether a failure is caused primarily by incorrect coarse routing or by confusion within the negative branch after correct routing.

Overall, the proposed methodology treats Vietnamese SER not only as a classification problem, but also as a reliability-sensitive decision component whose behavior should be evaluated at both aggregate and worst-case levels.

4 Experimental Setup

Our experiments are designed to answer three questions:

- **RQ1.** How do imbalance-aware training strategies affect aggregate performance and class-level reliability in one-stage Vietnamese SER?
- **RQ2.** Does hierarchical inference improve auditability and end-to-end reliability over flat classification?
- **RQ3.** Do the proposed reliability-aware components improve hierarchical SER beyond standard imbalance mitigation?

We conduct experiments on **SentiV**, a Vietnamese speech emotion dataset annotated from the original **VietSpeech** corpus [11,12]. All models use the same pretrained Vietnamese wav2vec2 backbone, fixed stratified train/validation/test split, preprocessing pipeline, and optimization protocol. Model selection is performed on the validation split, and final comparison is reported on the held-out test set.

We evaluate four groups of systems. Group A contains a one-stage six-class baseline trained with standard cross-entropy. Groups B and C examine imbalance-aware variants in the flat setting through loss-level and data-level interventions, including class weighting, focal loss, and augmentation. Group D contains the full one-stage recipe. Group E contains hierarchical variants for analyzing routing quality, branch-level discrimination, and encoder transfer.

Table 1. Compared system groups and their experimental roles.

Group	Systems	Purpose
A	One-stage classifier with standard cross-entropy	Flat six-class baseline
B	One-stage + weighted CE; one-stage + focal loss; one-stage + focal loss + weights	Loss-level imbalance mitigation
C	One-stage + uniform augmentation; one-stage + minority-aware augmentation	Data-level imbalance mitigation
D	One-stage + focal loss + weights + minority-aware augmentation + LLRD	Full one-stage recipe
E	Two-stage + CE; two-stage full recipe; two-stage without Stage-1 initialization	Hierarchical variants and encoder transfer
F	Two-stage + soft inference; two-stage + routing-aware regularization; two-stage + silver-aware smoothing	Reliability-aware ablations

5 Results and Analysis

5.1 One-Stage Results

Table 2 reports the one-stage results. The standard CE baseline remains the strongest model in aggregate terms, achieving **0.7529** Accuracy and **0.7375**

Macro-F1. Neither weighting, focal loss, nor augmentation surpasses this baseline on overall recognition quality. This suggests that, under the current Vietnamese SER setting, imbalance-aware training does not automatically improve flat six-class recognition.

When the evaluation shifts from average performance to class-level reliability, the picture changes. As shown in Table 3, focal loss with class weights yields the lowest recall disparity (**0.1327** compared with 0.1852 for the CE baseline), but this comes at a noticeable cost in Macro-F1 (0.7172 vs. 0.7375). The full one-stage recipe achieves the highest Macro-Recall (**0.7393**), yet it does not improve either worst-class recall or recall disparity relative to the CE baseline. Overall, the flat setting exhibits a clear balance-performance trade-off: methods that make recall more even across classes tend to weaken aggregate recognition quality, while the strongest aggregate model is not the most balanced one.

Table 2. Test-set performance of the one-stage models. Higher is better for all metrics.

Model	Acc.	Macro-F1	Macro-P	Macro-R
One-stage + CE	0.7529	0.7375	0.7416	0.7346
One-stage + Weighted CE	0.7483	0.7348	0.7374	0.7343
One-stage + Focal Loss	0.7363	0.7187	0.7229	0.7152
One-stage + Focal Loss + Weights	0.7283	0.7172	0.7174	0.7190
One-stage + CE + Uniform Aug.	0.7504	0.7331	0.7391	0.7292
One-stage + CE + Minority-aware Aug.	0.7511	0.7335	0.7408	0.7292
One-stage Full Recipe	0.7393	0.7302	0.7311	0.7393

Table 3. Fairness-oriented recall summary for the one-stage models on the test set. Higher is better for worst-class recall; lower is better for recall disparity.

Model	Worst-class recall	Recall disparity
One-stage + CE	0.6617	0.1852
One-stage + Weighted CE	0.6600	0.1609
One-stage + Focal Loss	0.6203	0.2201
One-stage + Focal Loss + Weights	0.6433	0.1327
One-stage + CE + Uniform Aug.	0.6437	0.2082
One-stage + CE + Minority-aware Aug.	0.6430	0.2152
One-stage Full Recipe	0.6473	0.1954

5.2 Hierarchical Results

The hierarchical setting reveals an even sharper trade-off. Table 4 shows that the full recipe substantially improves Stage 2, increasing Macro-F1 from 0.8191 to **0.8439** and Macro-Recall from 0.8250 to **0.8526**. However, these gains are

Table 4. Stage-wise performance of the hierarchical models.

Model	S1 Acc.	S1 F1	S1 R	S2 Acc.	S2 F1	S2 R
Two-stage + CE	0.8055	0.7361	0.7378	0.8334	0.8191	0.8250
Two-stage Full Recipe	0.7310	0.6959	0.7873	0.8564	0.8439	0.8526
Two-stage w/o S1 Init.	0.7391	0.7026	0.7901	0.8528	0.8388	0.8461

accompanied by a clear deterioration in Stage 1 routing, whose Accuracy drops from **0.8055** to 0.7310 and whose Macro-F1 drops from **0.7361** to 0.6959. This already indicates that stronger within-branch discrimination is insufficient when the upstream routing stage becomes less reliable.

The end-to-end analysis in Table 5 confirms this conclusion. Although the full hierarchical recipe improves recall for *Enjoyment* and *Neutral*, it becomes substantially less reliable on the negative emotions handled by Stage 2. Its worst-class recall falls from **0.6210** to 0.5276, while recall disparity rises sharply from **0.1345** to 0.3834. Therefore, stronger intermediate performance does not necessarily imply a better final system once routing errors are taken into account. This result directly motivates the responsibility-oriented view adopted in this paper: reliable SER systems should be judged not only by branch-level or average performance, but by their end-to-end behavior under the weakest classes.

Table 5. Reconstructed end-to-end six-class recall and fairness summary for the hierarchical models on the test set. End-to-end recalls are estimated from Stage 1 routing recall and Stage 2 conditional recall.

Model	Enjoy.	Neutral	Anger	Fear	Sad.	Surp.	Worst	Disp.
Two-stage + CE	0.6210	0.7233	0.7480	0.6637	0.7007	0.7555	0.6210	0.1345
Two-stage Full Recipe	0.7720	0.9110	0.6090	0.5276	0.5989	0.5792	0.5276	0.3834

5.3 Stage-Wise Error Attribution

To understand why some hierarchical variants fail end-to-end, Table 6 decomposes negative-emotion errors into routing and within-branch components. In the CE-based hierarchical model, routing error remains relatively limited (0.131 across negative classes), and end-to-end failures are shared between gating and branch confusion. In contrast, the full recipe suffers from much larger routing error (0.321), even though branch errors are generally smaller. This means that the main bottleneck shifts from within-branch discrimination to incorrect coarse routing. This decomposition is central to the paper’s trustworthy-AI perspective. A hierarchical model may appear stronger when judged only by Stage 2 metrics, yet still be less reliable as a complete decision component if routing remains fragile. The stage-wise view therefore provides evidence that aggregate metrics alone cannot reveal, and it explains why branch-level gains must be interpreted with caution in decision-support settings.

Table 6. Decomposition of negative-emotion end-to-end errors into routing and within-branch components.

Model	Class	Branch R	E2E R	Gate Err.	Branch Err.
Two-stage + CE	Anger	0.861	0.748	0.131	0.121
	Fear	0.764	0.664	0.131	0.205
	Sadness	0.806	0.701	0.131	0.168
	Surprise	0.869	0.756	0.131	0.114
Two-stage + Full	Anger	0.897	0.609	0.321	0.070
	Fear	0.777	0.528	0.321	0.151
	Sadness	0.882	0.599	0.321	0.080
	Surprise	0.853	0.579	0.321	0.100

5.4 Discussion of Findings

The experiments support three main findings. First, in the one-stage setting, imbalance-aware training mainly changes the balance-performance trade-off rather than improving the strongest overall model. Second, in the hierarchical setting, better Stage 2 performance does not ensure better end-to-end reliability because routing remains a major source of error. Third, the value of the proposed reliability-aware components should be judged primarily by end-to-end worst-class behavior and routing robustness, not only by branch-level metrics. Overall, Vietnamese SER under class imbalance and silver-label uncertainty should be evaluated as a reliability-sensitive decision problem, not only through average benchmark scores. In this setting, the main benefit of the hierarchical formulation is transparent diagnosis of where reliability is gained or lost.

6 Conclusion

Vietnamese SER should not be evaluated solely through aggregate metrics, especially under class imbalance and silver-label uncertainty. In the one-stage setting, the CE baseline achieves the strongest overall performance, while fairness-aware variants mainly affect class-level coverage rather than improving Accuracy or Macro-F1. Although some strategies increase Macro-Recall, they do not consistently improve balanced recognition across emotions. The hierarchical setting reveals a different trade-off. Fairness-aware training strengthens the negative-emotion branch, but these gains do not consistently translate into better end-to-end performance because Stage 1 routing remains a key source of error. This shows that improving an intermediate component is not sufficient to improve the reliability of the full system.

Overall, the results highlight a practical implication: Vietnamese SER under class imbalance and label uncertainty should be evaluated at both the component and end-to-end levels, with emphasis on worst-case behavior rather than average performance alone.

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